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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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Starlight — FreshPlate

1. Team & Project Overview

Team Name	Starlight
Selected Domain	Domain 2 - ALPHA RED
Selected SDG Track (1–17)	12 — Responsible Consumption and Production; Focus: Target 12.3
Project / Solution Name	FreshPlate
Name (Leader)	WANG HAOTONG
Email & Phone (Leader)	EEE2509044@xmu.edu.my | +60176130343
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small Chinese restaurants in Malaysia need to turn soon-to-expire stock into feasible prep portions while protecting existing orders. Delivery notes and recipes arrive as text or images; ingredients are shared across dishes. Separate calculations risk over-allocation. Owners need approval control and traceable batch consumption.

UNEP estimates that global food waste reached 1.05 billion tonnes in 2022, including inedible parts; food services accounted for 28% [1]. FreshPlate focuses on the food-service opportunity within SDG 12.3: maintaining usable batch information and turning it into controlled preparation actions [2].

Easier entry supports reliable batch records. Shared allocation, reservation protection and recorded completion connect those records to kitchen decisions and help owners manage daily preparation systematically.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FreshPlate is deployed: view/update inventory → generate prep plans → approve → complete → track consumption. Gemini extracts stock or recipes from text or one JPG/PNG. Users compare the source with an editable preview, resolve quantity, unit, ingredient and date issues, then confirm saving. Manual entry, custom ingredients and recipes normalized per serving are supported.

Serving quantities and batch allocation are calculated by a deterministic heuristic using one shared inventory budget. Eligible batches expiring today through two days ahead receive priority, and earlier-expiring batches are allocated first. Whole servings respect owner-set prep caps and existing-order reservations.

Approval reserves the saved plan snapshot without deducting stock. Completion deducts its allocated batches once and releases only that plan's reservations; cancellation releases them without consumption. Activity Log links the action, dishes and stock changes. Inventory, recipes and snapshots persist in the current browser's versioned localStorage.

Vercel hosts the interface and extraction function. Online functional acceptance is complete. I have completed all mandatory GenAI, Database and Cloud Deployment workshops; the implementation applies API integration and cloud hosting skills. Replit ran the identical planner and produced 61/45 simulated plan servings.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

React 19.3.1, Vite 8.3.5, JavaScript/CSS and Node.js implement the prototype. A Vercel Node.js Function exposes POST /api/extract; server-only GEMINI_API_KEY and GEMINI_MODEL configure Gemini. Structured output and field validation feed the same reviewed import path. Browser localStorage stores confirmed data, snapshots and activity.

Extraction: text/image → server function → Gemini draft → edit/validate → confirm → state.
Preparation: state → shared planner → Draft → Approved → Completed/Cancelled → inventory + Activity Log.

Replit sponsor use: run the same core module with default caps and a zero Tomato & Egg Stir-fry cap, producing 61 and 45 simulated servings, respectively.

Prototype: https://fresh-plate-psi.vercel.app
Repository: https://github.com/ShatteredDreams819726/FreshPlate
Replit: https://replit.com/@KarenAijo/FreshPlate-Planner-Demo

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Starting from receiving deliveries, recording stock and arranging prep in small restaurants, I identified two needs: easier data entry and actionable use of priority stock. FreshPlate combines Gemini extraction, editable review and batch-based planning. Repeated workflow checks led to larger text, simpler prompts, date pickers and collapsible details; automated checks and online acceptance verified allocation, reservation protection and stock updates.

Phase 2 proposes about three operating days at one cooperating restaurant, subject to scheduling and access. **Day 1**: time manual stock/recipe entry and record existing prep practices and edible ingredient/unsold-food discards. **Days 2–3**: use FreshPlate for extraction, review, import, planning and execution logs; record total verified entry time, corrections and user feedback. Weigh discards daily and record servings, purchases and menu changes.

Compare matched entry tasks and edible discard kg per 100 servings before/after, alongside priority-batch consumption. Results will assess efficiency and explore short-term waste trends, informing broader testing.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s) **Identify the exact UN SDG sub-target your project addresses e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."**	SDG 12.3 aims by 2030 to halve per-capita global food waste at retail/consumer levels and reduce supply-chain food losses [2]. FreshPlate contributes in food preparation through maintainable batch records, earlier-expiry allocation and traceable preparation.
Current Problem Baseline **What is the starting condition of the problem, expressed as a number? e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."**	Industry context: UNEP's global 2022 estimate is 1.05 billion tonnes, including inedible parts; 28% came from food services [1]. Restaurant baseline: Day 1 of the proposed pilot will measure manual-entry time and edible food discards under existing practices.
Core Measurable KPI **The specific unit you will track e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.**	**Core KPIs:** • **Data entry time reduction (%):** Compare manual and AI-assisted entry times for inventory and recipes separately. Timing includes AI processing, human review, and corrections until accurate data is saved. Reduction = (manual entry time − AI-assisted entry time) ÷ manual entry time × 100%. • **Food waste reduction (%):** Compare the weight of edible food discarded per 100 meals served before and after using FreshPlate, measured in kilograms and including raw ingredients and unsold prepared food. **Supporting metrics:** Track AI field extraction accuracy and the number of fields requiring manual correction per entry to assess extraction quality and ease of use.

Metric Category	Your Response
Phase 2 Impact Target **Your realistic, percentage-based target by the final prototype presentation e.g., "20% improvement in computational resource use."**	**Proposed Phase 2 targets**: 20% lower verified AI-assisted entry time than manual entry, assessed separately for stock and recipes, and 30% lower edible discard kg per 100 servings during the approximately three-day restaurant pilot.The short before/after observation explores waste trends.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No. I have not reused personal code, design or research created before the hackathon. Third-party components and AI-assisted development are disclosed in 6.2.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / React DOM 19.3.1; Vite 8.3.5; React plugin 6.1.2	Code / frameworks and build tools	MIT; retain copyright and license notices [10].
Scheduler, Rolldown, PostCSS and supporting packages	Automatically installed dependencies	MIT; bundled third-party notices also apply [10].
Lightning CSS and native binding; detect-libc; picocolors; source-map-js	Indirect code dependencies	MPL-2.0; Apache-2.0; ISC; BSD-3-Clause, respectively [10].
Node.js	Code / runtime	MIT and bundled third-party notices.
Google Gemini API	Pretrained model service / extraction	Google API and Gemini service terms [3].

Name	Type (code/dataset/model)	License
ChatGPT / Codex	AI-assisted development	Applicable OpenAI terms [4].
Replit; Vercel; GitHub	Execution; hosting; repository services	Respective service terms [5–7].
Visual Studio Code	Development editor	Microsoft product license; source code has a separate MIT license [8].
Unicode emoji; system fonts; inline SVG; official template artwork	UI symbols / fonts / document branding	Platform/font rights and organizer artwork apply; not uniformly MIT [9].

The package-lock and installed-license audit also covers @oxc-project/types, @rolldown/pluginutils, the Rolldown Windows binding, fdir, nanoid, picomatch and tinyglobby. Vite's LICENSE.md and Rolldown's THIRD-PARTY-LICENSE contain additional bundled notices, including BSD-2-Clause, CC0-1.0, ISC and MIT components. Preserve the notices and applicable source requirements for redistributed components.

Application icons use Unicode emoji and inline SVG; the CSS selects Segoe UI/Arial.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

No employment, internship or consulting IP conflict applies to this project.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Leader name: **WANG HAOTONG**

Electronic signature:_____ 王浩潼 _____

Actual signing date: **2026-10-23**

Appendix

[1] UNEP. World squanders over 1 billion meals a day — UN report. 27 March 2024. Global 2022 estimates, including inedible parts. Official UNEP release

[2] United Nations DESA. Goal 12, Target 12.3 and indicator 12.3.1(b). Official SDG targets

[3] Google. Gemini API Additional Terms of Service. Hosted model/API service terms. Gemini terms

[4] OpenAI. Terms of Use. ChatGPT/Codex use is subject to the applicable service and account terms. OpenAI terms

[5] Replit. Terms of Service. Replit terms

[6] Vercel. Terms of Service. Vercel terms

[7] GitHub. Terms of Service. GitHub terms

[8] Microsoft. Visual Studio Code software license. The distributed product and its MIT-licensed source are distinguished. VS Code product license

[9] Microsoft. Font redistribution FAQ and OpenType OS/2 embedding permissions. The template's five embedded font parts carry editable-embedding flags (0x0008). Font rights Embedding flags

[10] Dependency license evidence: package-lock.json and the installed packages' LICENSE/NOTICE files; Vite LICENSE.md and Rolldown THIRD-PARTY-LICENSE. Primary project license references: React license Vite license